RETAIL SALES & BUSINESS PERFORMANCE ANALYTICS

End-to-end data analytics project using Python, SQL and Power BI

Project Report

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1. Executive Summary

This project analyzes the Sample Superstore retail dataset to understand sales, profitability, customer segments, product categories, shipping modes and regional performance. Python was used for data preparation and exploratory analysis, MySQL for business queries, and Power BI for an interactive dashboard. The resulting report brings key performance indicators and visual analysis together to support data-driven business decisions.

2. Project Objectives

• Measure overall sales, profit, order volume, quantity and profit margin.

• Compare sales and profit across product categories and sub-categories.

• Understand the contribution of customer segments and shipping modes.

• Compare sales performance across regions and states.

• Explore sales performance over time using order dates.

• Present findings in an interactive Power BI dashboard.

3. Dataset Overview

The project uses the Sample Superstore dataset. It contains order-level retail information such as Order ID, Order Date, Ship Date, Ship Mode, customer details, geographic fields, product category and sub-category, Sales, Quantity, Discount and Profit. The original dataset contained 9,994 rows; the cleaned file used in the analysis contains 9,694 rows.

Data preparation note: The project workflow included duplicate removal, date and numeric type conversion, and removal of rows missing critical fields before exporting cleaned CSV and Excel files.

4. Tools and Technologies

Tool	Purpose
Python (Pandas, NumPy)	Data cleaning, transformation and KPI calculations
Jupyter Notebook	Interactive Python analysis and documentation
MySQL	Relational storage and SQL-based business analysis
Power BI Desktop	Interactive dashboard and data visualization

Tool	Purpose
Excel / CSV	Cleaned dataset exchange and Power BI import

5. Key Performance Indicators

Metric	Result
Total Sales	2,297,200.86
Total Profit	286,397.02
Total Orders	5,009
Total Quantity	37,873
Profit Margin	12.47%

Profit margin is calculated as Total Profit divided by Total Sales, multiplied by 100. Displayed Power BI card values are rounded for readability.

6. SQL Analysis

The SQL analysis file contains queries for overall sales, profit and orders; sales and profit by category; sales by region; top customers by sales; profit by sub-category; monthly sales; sales by customer segment; sales by shipping mode; and overall profit margin. These queries provide reusable answers to common retail business questions.

7. Power BI Dashboard

The dashboard includes KPI cards for Total Sales, Total Profit, Total Quantity and Profit Margin, along with a separate Total Orders card. Visuals show sales by category, geographical sales distribution, sales by customer segment, profit by sub-category, orders by shipping mode, year-wise sales trend, sales by region and profit by category. Category, Region and Order Date slicers allow interactive filtering.

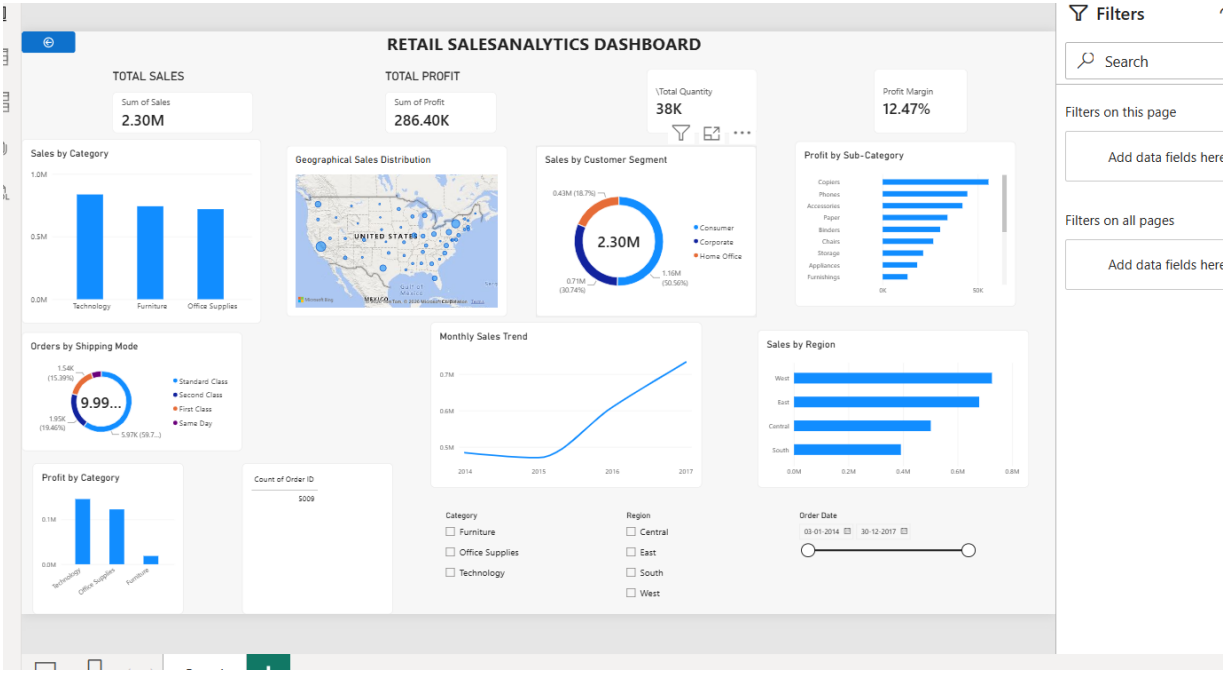


Figure 1. Power BI dashboard screenshot captured during project development.

8. Key Observations

• Technology is the highest-sales category in the dashboard comparison.

• The West region records the highest sales among the four regions shown.

• The Consumer segment contributes the largest share of sales.

• The overall profit margin is approximately 12.47%.

• The year-wise trend chart shows sales increasing from 2015 through 2017.

9. Conclusion

This project demonstrates a complete analytics workflow, from preparing raw retail data to querying business metrics and presenting findings in an interactive dashboard. It combines Python, SQL and Power BI to make retail performance easier to explore. The project can be extended with deeper discount analysis, customer retention analysis, forecasting and more detailed product-level profitability investigation.

10. Project Deliverables

• Cleaned dataset in CSV and Excel formats.

• Jupyter Notebook and exported Python script.

• MySQL database table and SQL analysis script.

• Power BI dashboard (.pbix) and screenshot.

• This project report in PDF format.